

Linear-Error Clique Partitions of Split Graphs via Structured Triangle Packing

Juan Pablo Traverso Gianini

Independent researcher, Santiago, Chile

jtraverso@gmail.com

ORCID: 0009-0003-6068-4096

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Paper III in the series

Author manuscript: v0.9.12 editorial review candidate. The mathematical statements, hypotheses, constants, displayed identities, and proofs in Sections 2–9 and Appendices A–D are unchanged from v0.9.7, apart from cross-reference labels updated in v0.9.10. Version 1.0 is reserved for the first public preprint.

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Status: editorial preprint candidate. The author-supplied Lean 4 / Mathlib v4.28.0 build and axiom report records a zero-`sorry` compile for the proved core, with no `admit`, `native_decide`, or `unsafe`. Its trusted project layer consists of exactly the two external asymptotic inputs of Section 2.4, retained as the named axioms **AX1** (Haxell–Rödl/Yuster) and **AX2** (Dross + Barber–Kühn–Lo–Osthus). The reported axiom footprint of Theorem 1.1 is `propext`, `Classical.choice`, `Quot.sound`, **AX1**, and **AX2**, and the report records no other project axiom declaration. The public release commit, an independent reproduction of the build, human peer review, and the prior-art assessment remain pending; final verification wording is tied to the frozen release. See Section 11.6 for the full perimeter.

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Abstract

Let $G = (K \cup I, E)$ be a split graph on n vertices, with K a clique and I an independent set. We prove that there is an absolute constant C such that

$$|E(G)| - 2\nu_3(G) \leq \frac{n^2}{6} + Cn,$$

where $\nu_3(G)$ is the maximum number of pairwise edge-disjoint triangles in G . Consequently,

$$\text{cp}(G) \leq \frac{n^2}{6} + Cn$$

for every split graph G , where $\text{cp}(G)$ is the minimum number of cliques whose edge sets partition $E(G)$. This strengthens the asymptotic $(1/6 + o(1))n^2$ bound to the correct linear-error scale. The leading constant is sharp.

The proof separates three regimes according to $\alpha = |I|/|K|$. In the bulk regime, an exact four-orbit linear program for a common-neighborhood profile yields a uniform quadratic fractional margin; the Haxell–Rödl theorem then absorbs the subquadratic integrality loss. When $\alpha \rightarrow 0$, large edge-disjoint matchings anchored at the independent vertices leave an almost complete,

triangle-divisible clique residual, which is decomposed exactly by dense graph decomposition results. When $\alpha \rightarrow 2$, averaged factorization closes the short corridor, while a double-factor polarization inequality and a shifted-center gain-completion argument close the remaining mesoscopic corridor.

The proof does not establish the stronger universal estimate $\nu_3^*(G) - \nu_3(G) = O(n)$. Away from the extremal corridor, a potentially superlinear integrality gap is absorbed by quadratic fractional slack.

All corridor-specific mechanisms are proved in the paper; in particular, the list edge coloring ingredient is established from first principles in Appendix D. Beyond standard classical results used throughout graph theory, the only asymptotic external inputs are the two theorems confined to the bulk and sparse regimes. The near-extremal corridor theorem (Proposition 10.5) is elementary and effective.

Keywords: clique partition; split graph; triangle packing; fractional packing; factorization; polarization; list edge coloring; graph decomposition.

1. Introduction

1.1 Clique partitions of split graphs

A **clique partition** of a graph G is a family of complete subgraphs whose edge sets partition $E(G)$. Cliques of order two are allowed. The minimum size of such a family is denoted by $\text{cp}(G)$.

The chordal clique-partition problem of Erdős, Ordman, and Zalcstein asks for the correct extremal order of $\text{cp}(G)$ on chordal graphs [6]. Their complete-split (indeed threshold) construction shows that the coefficient $1/6$ is unavoidable. Split graphs form a natural and important subclass of chordal graphs: a graph is split if its vertex set can be partitioned into a clique and an independent set. Earlier work of Chen, Erdős, and Ordman established a sharper asymptotic upper bound for this subclass [3].

This is the third paper in the series on the Erdős–Ordman–Zalcstein clique-partition problem (Erdős Problem #81). Paper I proves a finite fractional split-graph bound, $|E(G)| - 2\nu_3^*(G) \leq n^2/6 + n$ [13]. Paper II identifies the complete-split terminal family and determines the exact finite maximum of the fractional cover functional over chordal graphs [8]. Papers I and II are fractional and have frozen Lean 4 releases. Paper IV of the series develops a general transfer and rounding interface for integral applications; it is logically independent of the present proof. The present paper is the first integral installment: it strengthens the split-graph estimate to the linear-error scale $n^2/6 + O(n)$.

Paper II of the series determines the fractional extremal problem exactly. For every n ,

$$\max_{\substack{|V(G)|=n \\ G \text{ chordal}}} (|E(G)| - 2\nu_3^*(G)) = \max_{\substack{|V(G)|=n \\ G \text{ split}}} (|E(G)| - 2\nu_3^*(G)) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.$$

It follows from the dense-graph fractional-to-integral packing theorems of Haxell–Rödl and Yuster [7,9] that every chordal graph satisfies

$$\text{cp}(G) \leq \left(\frac{1}{6} + o(1) \right) n^2.$$

The present paper addresses a different and genuinely second-order question: it strengthens the integral bound to $n^2/6 + O(n)$ for arbitrary split graphs.

1.2 Triangle packings and the linear-error target

Let $\nu_3(G)$ be the maximum number of pairwise edge-disjoint triangles in G . Every such packing gives a clique partition: keep every packed triangle and use one K_2 for each uncovered edge. Hence

$$\text{cp}(G) \leq |E(G)| - 2\nu_3(G). \quad (1.1)$$

The central theorem of this paper is an integral estimate at the correct second-order scale.

Theorem 1.1 — Linear-error split-graph theorem

There exists an absolute constant C such that every split graph G on n vertices satisfies

$$|E(G)| - 2\nu_3(G) \leq \frac{n^2}{6} + Cn. \quad (1.2)$$

Combining (1.1) and Theorem 1.1 gives the clique-partition statement.

Corollary 1.2 — Linear-error clique partition bound

There exists an absolute constant C such that every split graph G on n vertices satisfies

$$\text{cp}(G) \leq \frac{n^2}{6} + Cn. \quad (1.3)$$

The Erdős–Ordman–Zalcstein complete-split (indeed threshold) family $K_p \vee \overline{K}_{2p}$, with $n = 3p$, shows that the leading constant $1/6$ is sharp and that a linear term naturally appears.

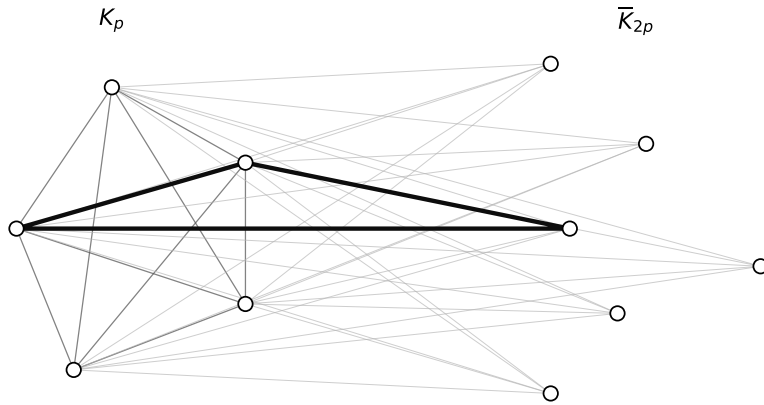


Figure 1: The complete-split graph $K_p \vee \overline{K}_{2p}$. Every edge between the two parts is present, and the independent side has no internal edges. Since every triangle contains a clique edge, an edge-disjoint triangle packing uses at most $\binom{p}{2}$ triangles; see Section 10.2. The figure is illustrative and is not used as a premise of the proof.

1.3 Localization of hypothetical obstructions

Proposition 1.3 — Localization of hypothetical obstructions

Suppose, for contradiction, that no absolute linear-error bound holds, and for each positive integer k choose a minimum-order split graph G_k satisfying

$$|E(G_k)| - 2\nu_3(G_k) > \frac{|V(G_k)|^2}{6} + k|V(G_k)|.$$

After passing to a subsequence, writing $|K_k| = p_k$, $|I_k| = q_k$, every such hypothetical sequence must satisfy

$$\frac{q_k}{p_k} \longrightarrow 2.$$

Moreover, if $q_k = 2p_k - s_k$, then the bulk and short-corridor estimates force

$$\sqrt{p_k} \ll s_k = o(p_k).$$

Thus any obstruction to a linear error term would have to concentrate in the mesoscopic near-extremal corridor. The high/low-dispersion dichotomy in Section 9 excludes that corridor.

1.4 What the theorem does not prove

Let $\nu_3^*(G)$ denote the fractional triangle-packing number. The present proof does **not** establish

$$\nu_3^*(G) - \nu_3(G) = O(n) \tag{1.4}$$

uniformly over split graphs.

Indeed, write

$$\Gamma(G) = \nu_3^*(G) - \nu_3(G)$$

and

$$S(G) = \frac{n^2}{6} - (|E(G)| - 2\nu_3^*(G)).$$

Then

$$|E(G)| - 2\nu_3(G) = \frac{n^2}{6} - S(G) + 2\Gamma(G). \tag{1.5}$$

Theorem 1.1 only requires

$$2\Gamma(G) \leq S(G) + O(n).$$

In the bulk regime, $S(G)$ is quadratic and absorbs the general $o(n^2)$ integrality loss. Thus the universal linear integrality-gap problem remains open.

1.5 Proof architecture

Write

$$|K| = p, \quad |I| = q, \quad \alpha = \frac{q}{p}.$$

The ratio $\alpha = q/p$ determines which source of slack is available. We first dispose of $q \geq 2p - 1$ by direct averaged factorization, and may then assume $0 \leq \alpha < 2$. A hypothetical sequence of counterexamples has a subsequence in one of three regimes.

1. **Bulk:** α stays away from both 0 and 2. An exact common-profile LP and a fractional cloning argument produce quadratic fractional slack. Haxell–Rödl converts the fractional packing into an integral one with only $o(n^2)$ loss.
2. **Sparse independent side:** $\alpha \rightarrow 0$. Each independent vertex anchors a large matching inside its neighborhood. The residual clique is almost complete. After deleting $O(p)$ edges to correct divisibility, it admits an exact triangle decomposition.
3. **Near-extremal corridor:** $\alpha \rightarrow 2$. Write $q = 2p - s$. Averaged factorization closes $s = O(\sqrt{p})$. For $\sqrt{p} \ll s = o(p)$, high dispersion is paid by a double-factor polarization term, while low dispersion implies closeness to a common center and is handled by shifted-center gain completion.

Table 1. The three proof regimes. The near-extremal corridor is elementary and effective (Proposition 10.5); the other two regimes each invoke exactly one external asymptotic input.

Regime	Condition on $\alpha = q/p$	Source of slack	External input	Sections
Sparse independent side	$\alpha \rightarrow 0$	dense clique residual admits an exact triangle decomposition	Theorem 2.3 (Dross–BKLO)	§8
Bulk	α bounded away from 0 and 2	quadratic fractional margin (common-profile LP + fractional cloning)	Theorem 2.1 (Haxell–Rödl/Yuster)	§3–4, §9.1
Near-extremal corridor	$\alpha \rightarrow 2, q = 2p - s$	averaged factorization, double-factor polarization, shifted-center gain completion	none (elementary/effective)	§5–7, §9.2–9.3, §10.5

The division is not cosmetic. In the bulk there is enough quadratic fractional slack to absorb a general $o(n^2)$ rounding loss. At the two ends that slack vanishes, so the proof replaces generic rounding by constructions adapted to the geometry of the split presentation.

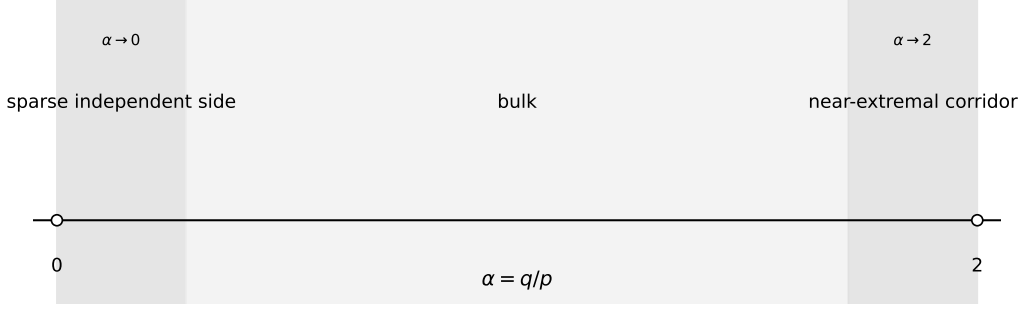


Figure 2: The three parameter regimes used in the proof. The principal corridor thresholds are shown; the complete hypotheses are stated in Sections 4, 8, and 9. The endpoint regimes are asymptotic rather than hard subintervals of $[0, 2]$.

1.6 Supplementary computational audits

The accompanying closure package contains exact-arithmetic and finite-instance regression tests for selected identities and quantitative inequalities used in the proof.

Every audited statement is proved analytically in the manuscript. No computation, script, finite enumeration, or solver output is a logical premise of Theorem 1.1. The supplementary material is provided solely for independent checking, regression testing, and reproducibility.

1.7 Organization

Section 2 fixes notation and external inputs. Section 3 solves the common-profile LP. Section 4 proves the fractional cloning bound and the global fractional margin. Section 5 develops averaged and double-factor rounding. Section 6 proves polarization. Section 7 proves shifted-center gain completion. Section 8 handles the sparse-independent regime. Section 9 assembles the three regimes. Section 10 records corollaries, Section 11 discusses the proof and its limitations, and Section 12 describes future uses and open directions. The appendices contain algebraic details, the divisibility correction, reproducibility material, and a self-contained proof of the list edge coloring theorem (Appendix D).

2. Preliminaries

2.1 Split notation

Throughout,

$$G = (K \sqcup I, E)$$

is a split graph, where K is a clique of size p and $I = \{v_1, \dots, v_q\}$ is independent.

For $v_i \in I$, write

$$N_i = N(v_i) \cap K, \quad d_i = |N_i|, \quad S_i = K \setminus N_i, \quad m_i = |S_i|.$$

Set

$$M = \sum_i m_i, \quad S_2 = \sum_i m_i^2.$$

When q is close to $2p$, write

$$q = 2p - s. \tag{2.1}$$

2.2 Triangle packings

A fractional triangle packing is a nonnegative weight assignment to the triangles of G such that the total weight through every edge is at most one. Its maximum value is $\nu_3^*(G)$.

The dual is a fractional triangle cover: a nonnegative weight assignment to edges such that every triangle receives total edge weight at least one. LP duality gives equality between the two optimum values.

Throughout we write

$$\Phi(G) := |E(G)| - 2\nu_3(G)$$

for the integral triangle-packing defect; by (1.1), $\text{cp}(G) \leq \Phi(G)$.

2.3 Complete graph factorizations

Adopt the boundary conventions

$$\chi'(K_0) = \chi'(K_1) = 0.$$

For $t \geq 2$, the edge-chromatic number of a complete graph is

$$\chi'(K_t) = \begin{cases} t-1, & t \text{ even,} \\ t, & t \text{ odd.} \end{cases} \tag{2.2}$$

Thus $E(K_t)$ decomposes into $\chi'(K_t)$ matchings. The conventions for $t \leq 1$ make the same language valid for empty edge sets.

2.4 External theorems

We use two external asymptotic inputs. Earlier versions cited a third, the list edge coloring theorem; the case required here is proved self-contained in Appendix D. Thus Sections 5–7 and Proposition 10.5 require no additional asymptotic input, though they use standard classical facts such as the edge coloring of complete graphs.

Theorem 2.1 — Haxell–Rödl/Yuster [7,9]

For every fixed graph H ,

$$\nu_H^*(G) - \nu_H(G) = o(|V(G)|^2)$$

uniformly over graphs G . We apply this with $H = K_3$.

Theorem 2.2 — List edge coloring (proved in Appendix D)

Let B be a bipartite graph with maximum degree $\Delta(B)$. If every edge e is assigned a list $L(e)$ with

$$|L(e)| \geq \Delta(B),$$

then B has a proper edge coloring choosing the color of each edge from its list.

This is the maximum-degree case of Galvin's theorem [10]. A self-contained proof (König coloring, stable pairings, and the kernel lemma) is given in Appendix D, so this is not an external dependency of the paper. The local refinement $|L(xy)| \geq \max\{d(x), d(y)\}$ of Borodin, Kostochka, and Woodall [2] is not needed: hypothesis (7.2) already bounds every list by the maximum degree of the gain graph.

Theorem 2.3 — Dense triangle decomposition

For every $\varepsilon > 0$, every sufficiently large triangle-divisible graph H with

$$\delta(H) \geq (0.9 + \varepsilon)|V(H)|$$

admits a triangle decomposition. This follows from Dross's fractional triangle decomposition theorem [5] (minimum degree $0.9v$ suffices fractionally) combined with the iterative-absorption theorem of Barber, Kühn, Lo, and Osthus [1], which converts the fractional threshold into an exact decomposition threshold for triangle-divisible graphs, up to ε and for sufficiently large order.

3. The Common-Profile Linear Program

For integers p, q, d , let $H(p, q, d)$ be the split graph with clique K , $|K| = p$, independent set I , $|I| = q$, and a fixed set $N \subseteq K$, $|N| = d$, such that every vertex of I has neighborhood N . Put $R = K \setminus N$ and $r = p - d$.

3.1 Symmetric cover variables

Averaging over permutations of N , R , and I , an optimal fractional triangle cover may be assumed constant on the four edge classes

$$E(N), \quad E(N, I), \quad E(N, R), \quad E(R).$$

Let their weights be a, b, c, e , respectively. The triangle constraints are

$$a + 2b \geq 1, \quad 3a \geq 1, \tag{3.1}$$

$$a + 2c \geq 1, \quad 2c + e \geq 1, \quad 3e \geq 1. \tag{3.2}$$

Constraints corresponding to empty triangle types may be omitted; the formula below remains valid for $p \geq 3$.

The objective is

$$\binom{d}{2}a + qdb + dr c + \binom{r}{2}e. \quad (3.3)$$

3.2 Exact solution

Theorem 3.1 — Common-profile formula

For $p \geq 3$ and $q \geq 1$,

$$\boxed{\nu_3^*(H(p, q, d)) = F(p, q, d),} \quad (3.4)$$

where

$$\boxed{F(p, q, d) = \min \left\{ \frac{\binom{p}{2} + qd}{3}, \binom{d}{2} + \binom{r}{2}, \binom{d}{2} + \frac{dr + \binom{r}{2}}{3} \right\}.} \quad (3.5)$$

Proof

At an optimum,

$$b = \frac{1-a}{2}, \quad c = \frac{1 - \min\{a, e\}}{2}, \quad \frac{1}{3} \leq a, e \leq 1.$$

If $a \leq e$, decreasing e to a cannot increase the objective. It therefore suffices to minimize on

$$\frac{1}{3} \leq e \leq a \leq 1.$$

The objective is affine on this triangle, so a minimum occurs at

$$(a, e) = \left(\frac{1}{3}, \frac{1}{3}\right), \quad (1, 1), \quad \left(1, \frac{1}{3}\right).$$

The corresponding values are precisely the three expressions in (3.5). Duality completes the proof. $\square$

When $q = 0$, the graph $H(p, 0, d)$ is simply K_p , independently of d , and the same formula follows from $\nu_3^*(K_p) = \binom{p}{2}/3$.

3.3 Interpretation of the three covers

The minimum in (3.5) is easier to remember through the three vertices of the reduced two-variable program. Each vertex represents a distinct way to pay for the triangle constraints.

Cover pattern	Reduced vertex (a, e)	Value	Geometric description
Uniform	$(1/3, 1/3)$	$((\binom{p}{2} + qd)/3)$	All relevant edge orbits are covered at the uniform fractional rate.
Separated	$(1, 1)$	$\binom{d}{2} + \binom{r}{2}$	The two clique blocks N and R are paid internally, with no weight on the crossing orbit.
Hot neighborhood	$(1, 1/3)$	$\binom{d}{2} + (dr + \binom{r}{2})/3$	The common neighborhood N is paid in full, while the residual clique structure remains fractional.

The labels are descriptive rather than additional definitions. In later calculations we refer to the corresponding first, second, and third branches of F . This finite trichotomy is the source of the global fractional margin.

4. Fractional Cloning and the Exact Fractional Margin

4.1 Fractional cloning

The fractional cloning operation (replication) used here is the cover-side analogue of the clone symmetrization used in Paper II. There, vertices or clone classes are copied pairwise to simplify a chordal graph. Here, one independent-set profile is copied q times in a single averaging step; the graph $H(p, q, d_i)$ records the resulting common profile.

Lemma 4.1 — Fractional cloning bound

For every split graph with $q \geq 1$,

$$\boxed{\nu_3^*(G) \geq \frac{1}{q} \sum_{i=1}^q F(p, q, d_i).} \quad (4.1)$$

Proof

Let y be any fractional triangle cover of G . Put

$$A = \sum_{e \in E(K)} y_e$$

and

$$B_i = \sum_{x \in N_i} y_{v_i x}.$$

Replace v_i by q independent clones, all with neighborhood N_i , and give each clone the incident weights of v_i . Together with the original clique-edge weights, this is a fractional cover of $H(p, q, d_i)$ of weight $A + qB_i$. Hence

$$A + qB_i \geq F(p, q, d_i).$$

Summing over i gives

$$q \left(A + \sum_i B_i \right) \geq \sum_i F(p, q, d_i).$$

Minimizing over covers proves the lemma. $\square$

4.2 The exact margin

Assume $0 < q \leq 2p$, set

$$\alpha = \frac{q}{p},$$

and define the packing threshold

$$T(G) = \frac{1}{2} \left(|E(G)| - \frac{(p+q)^2}{6} \right). \quad (4.2)$$

Define

$$\mu(\alpha) = \begin{cases} \alpha^2/12, & 0 \leq \alpha \leq 2/3, \\ (2-\alpha)^2/48, & 2/3 \leq \alpha \leq 2. \end{cases} \quad (4.3)$$

Theorem 4.2 — Unified fractional margin

$$\boxed{\nu_3^*(G) \geq T(G) + \mu(\alpha)p^2 - \frac{p}{4}.} \quad (4.4)$$

Proof

Let

$$C_\alpha = \frac{2 - 2\alpha - \alpha^2}{12}.$$

For each branch of $F(p, q, d)$, completion of squares gives

$$F(p, q, d) \geq \frac{qd}{2} + C_\alpha p^2 + \mu(\alpha)p^2 - \frac{p}{2}. \quad (4.5)$$

The three normalized residual minima are

$$\frac{\alpha^2}{12}, \quad \frac{(2-\alpha)^2}{48},$$

and

$$\begin{cases} \alpha(8 - 5\alpha)/48, & \alpha \leq 4/3, \\ (2 - \alpha)^2/12, & \alpha \geq 4/3. \end{cases}$$

The third is never below the minimum of the first two. Averaging (4.5) through Lemma 4.1 gives

$$\nu_3^*(G) \geq \frac{1}{2} \sum_i d_i + C_\alpha p^2 + \mu(\alpha) p^2 - \frac{p}{2}.$$

Since

$$T(G) = \frac{1}{2} \sum_i d_i + C_\alpha p^2 - \frac{p}{4},$$

we obtain (4.4). $\square$

4.3 Bulk consequence

If

$$\varepsilon \leq \alpha \leq 2 - \varepsilon,$$

then $\mu(\alpha) \geq c_\varepsilon > 0$. Theorem 4.2 and Haxell–Rödl imply

$$\nu_3(G) \geq T(G)$$

for all sufficiently large graphs in this regime. Hence

$$|E(G)| - 2\nu_3(G) \leq \frac{n^2}{6}. \tag{4.6}$$

5. Factorization Rounding Near $q = 2p$

Let

$$r_p = \chi'(K_p).$$

5.1 One-factor averaging

Lemma 5.1

If $q \geq r_p$, then

$$\boxed{\nu_3(G) \geq \frac{1}{q} \sum_i \binom{d_i}{2}}. \tag{5.1}$$

Proof

Factor K_p into r_p matchings. Assign the factors injectively and uniformly to vertices of I . In the factor assigned to v_i , retain only edges with both endpoints in N_i . The retained edges form valid edge-disjoint KKI triangles.

The expected number retained is

$$\frac{1}{q} \sum_i \sum_{j=1}^{r_p} |F_j \cap E(K[N_i])| = \frac{1}{q} \sum_i \binom{d_i}{2}.$$

Some assignment attains at least the expectation. $\square$

Writing $q = 2p - s$, substitution gives

$$\Phi(G) \leq \frac{n^2}{6} + \frac{p}{2} - \frac{s^2}{6} + \frac{(s-1)M - S_2}{q}. \quad (5.2)$$

Using $S_2 \geq M^2/q$ and maximizing the resulting parabola yields

$$\boxed{\Phi(G) \leq \frac{n^2}{6} + \frac{p}{2} + \frac{s^2 - 6s + 3}{12}}. \quad (5.3)$$

Thus $s = O(\sqrt{p})$ is closed with linear error.

5.2 Double-factor rounding

For a clique edge $e = xy$, let

$$b_e = |\{i : \{x, y\} \not\subseteq N_i\}|.$$

Put

$$h = \min\{r_p, q - r_p\}, \quad \delta = \frac{h}{r_p}.$$

Choose the set of h factors that receive two distinct independent vertices uniformly at random, give one independent vertex to every other factor, and assign the independent vertices to the resulting positions uniformly and injectively. All expectations below are taken over both random choices; some deterministic choice attains at least the expected gain.

Lemma 5.2 — Double-factor inequality

If $q \geq r_p$, then

$$\boxed{\Phi(G) \leq \frac{n^2}{6} + \frac{p}{2} - \frac{s^2}{6} + \frac{(s-1)M - S_2}{q} - \frac{2\delta V}{q(q-1)}}, \quad (5.4)$$

where

$$V = \sum_{e \in E(K)} b_e(q - b_e). \quad (5.5)$$

Proof

If the factor containing e receives one vertex, e is lost with probability b_e/q . If it receives two vertices, it is lost only when both are bad, with probability

$$\frac{b_e(b_e - 1)}{q(q - 1)}.$$

Thus the expected number U of lost clique edges is

$$U = \frac{1}{q} \sum_e b_e - \frac{\delta}{q(q - 1)} \sum_e b_e(q - b_e).$$

Because every factor is a matching, each retained edge may be assigned to one of its admitting vertices without creating an edge collision. Substitution gives (5.4). $\square$

6. Quantitative Polarization

For every i , define

$$\mathcal{B}_i = \{e \in E(K) : e \cap S_i \neq \emptyset\}.$$

Then

$$V = \sum_{i,j} |\mathcal{B}_i \setminus \mathcal{B}_j|. \quad (6.1)$$

Let

$$m = \max_i |S_i|.$$

Lemma 6.1 — Polarization inequality

If $2p - 3m - 1 \geq 0$, then

$$V \geq \frac{2p - 3m - 1}{4} \sum_{i,j} |S_i \triangle S_j|. \quad (6.2)$$

Proof

Put

$$a_{ij} = |S_i \setminus S_j|.$$

The edges in $\mathcal{B}_i \setminus \mathcal{B}_j$ are precisely the edges of $K_p - S_j$ meeting $S_i \setminus S_j$. Therefore

$$|\mathcal{B}_i \setminus \mathcal{B}_j| = \frac{a_{ij}(2(p - |S_j|) - a_{ij} - 1)}{2}.$$

Since $|S_j| \leq m$ and $a_{ij} \leq m$,

$$|\mathcal{B}_i \setminus \mathcal{B}_j| \geq \frac{2p - 3m - 1}{2} |S_i \setminus S_j|.$$

Sum over ordered pairs and use

$$\sum_{i,j} |S_i \setminus S_j| = \frac{1}{2} \sum_{i,j} |S_i \triangle S_j|.$$

This proves (6.2). $\square$

7. Shifted-Center Gain Completion

Fix $R \subseteq K$. Put

$$\rho = |R|, \quad Q = K \setminus R, \quad b = |Q|.$$

For each i , define

$$T_i = S_i \setminus R, \quad t_i = |T_i|,$$

and

$$G_i = R \setminus S_i, \quad g_i = |G_i|.$$

Set

$$A_R = \sum_i t_i, \quad A_{2,R} = \sum_i t_i^2, \quad B_R = \sum_i g_i.$$

Let

$$r_b = \chi'(K_b), \quad u = q - r_b.$$

Assume

$$b \geq 2, \quad q \geq r_b, \quad b \geq \chi'(K_\rho), \quad (7.1)$$

and

$$b - t_i \geq \max\{\rho, u\} \quad \text{for all } i. \quad (7.2)$$

Define

$$\theta_R = \frac{\max\{\rho - 1, 0\}}{b}$$

and

$$\kappa_R = 1 - 2(1 - \theta_R) \frac{u}{q}. \quad (7.3)$$

7.1 Packing inside Q

Reserve a set $U \subseteq I$ of u vertices; note that $|I \setminus U| = q - u = r_b$ exactly. Assign the remaining r_b vertices bijectively to a factorization of $K[Q]$.

For v_i , the number of unavailable edges of $K[Q]$ is

$$\beta_i = \binom{b}{2} - \binom{b - t_i}{2}. \quad (7.4)$$

For fixed U , averaging over assignments gives at least

$$\binom{b}{2} - \frac{1}{r_b} \sum_{i \notin U} \beta_i \quad (7.5)$$

edge-disjoint QQI triangles.

7.2 The gain graph

Construct a bipartite graph with parts U and R , joining v_i to r when $r \in G_i$. Assign the list

$$L(v_i r) = N_i \cap Q.$$

By (7.2), every list satisfies

$$|L(v_i r)| = b - t_i \geq \max\{\rho, u\} \geq \Delta,$$

where Δ is the maximum degree of the gain graph, since $d(v_i) = g_i \leq \rho$ and $d(r) \leq |U| = u$. Theorem 2.2, proved self-contained in Appendix D, gives a proper list edge coloring. If $v_i r$ receives color $z \in Q$, take the triangle $v_i r z$. This gives

$$B_U = \sum_{i \in U} g_i$$

edge-disjoint IRQ triangles.

7.3 Completion inside R

For each $z \in Q$, let $U_z \subseteq R$ be the vertices r for which rz was used by an IRQ triangle.

If $\rho \leq 1$, the graph $K[R]$ has no edges, so this completion contributes no RRQ triangles. Suppose therefore that $\rho \geq 2$. Factor $K[R]$, inject its factors into the colors $z \in Q$, and delete from the factor assigned to z every edge incident with U_z . Averaging over the injections loses at most

$$\frac{\rho-1}{b} \sum_z |U_z| = \theta_R B_U$$

edges. Hence, in all cases, we obtain at least

$$\binom{\rho}{2} - \theta_R B_U$$

additional RRQ triangles.

The three families QQI , IRQ , and RRQ are edge-disjoint. In particular, the forbidden-color deletion prevents an IRQ triangle and an RRQ triangle from sharing an edge rz .

7.4 The centered inequality

Choose U , $|U| = u$, maximizing

$$\sum_{i \in U} \left(\frac{2\beta_i}{r_b} + 2(1 - \theta_R)g_i \right).$$

The best u terms have sum at least u/q times the total. Using

$$2 \sum_i \beta_i = (2b-1)A_R - A_{2,R},$$

we obtain the principal local lemma.

Lemma 7.1 — Reserved-gain shifted-center inequality

Under (7.1)–(7.2),

$$\boxed{\begin{aligned} \Phi(G) &\leq \frac{n^2}{6} + \frac{p}{2} - \frac{s^2}{6} + s\rho - 2\rho^2 \\ &\quad + \kappa_R B_R + \frac{(s-2\rho-1)A_R - A_{2,R}}{q}. \end{aligned}} \tag{7.6}$$

8. The Sparse-Independent Regime

Assume $q = o(p)$. We prove the required bound directly and integrally.

8.1 Minimal-counterexample degree bound

In the global contradiction argument, a minimal counterexample with penalty kn satisfies, for every $v \in I$,

$$d(v) > \frac{2n-1}{6} + k. \quad (8.1)$$

Since $q = o(p)$, this implies eventually

$$d(v) \geq 2q + 2. \quad (8.2)$$

8.2 Successive matchings

Order $I = \{v_1, \dots, v_q\}$. Choose successively edge-disjoint matchings

$$M_i \subseteq K[N_i]$$

of size $\lfloor d_i/2 \rfloor$.

Before choosing M_i , each vertex has lost at most $i-1$ incident edges, so the available graph on N_i has minimum degree at least

$$d_i - i \geq d_i/2.$$

Dirac's theorem [4] supplies a Hamilton cycle and hence a matching of the required size.

Let

$$F = \bigcup_i M_i.$$

Then

$$|F| \geq \frac{1}{2} \sum_i d_i - \frac{q}{2}, \quad \Delta(F) \leq q. \quad (8.3)$$

The edges of M_i yield $|M_i|$ edge-disjoint KKI triangles with apex v_i .

8.3 Divisibility correction

The residual clique graph

$$R_0 = K_p - F$$

satisfies

$$\delta(R_0) \geq p - 1 - q = (1 - o(1))p. \quad (8.4)$$

In particular, for all sufficiently large members of the sequence, $\delta(R_0) > p/2$. Dirac's theorem [4] therefore gives a Hamilton cycle, and hence a Hamilton path

$$P = x_1 x_2 \cdots x_p$$

contained in R_0 . Let O be the set of odd-degree vertices of R_0 . By the handshaking lemma, $|O|$ is even. Define the path subgraph $J \subseteq P$ by

$$x_j x_{j+1} \in E(J) \iff |O \cap \{x_1, \dots, x_j\}| \text{ is odd.} \quad (8.5)$$

For an internal vertex x_j , the parity of its degree in J is the change in prefix parity between positions $j-1$ and j , and is therefore 1 precisely when $x_j \in O$. The same conclusion holds at the two endpoints because $|O|$ is even. Consequently,

$$\text{Odd}(J) = O, \quad |E(J)| \leq p-1, \quad \Delta(J) \leq 2. \quad (8.6)$$

Thus

$$R_1 := R_0 - J$$

has all degrees even, and

$$\delta(R_1) \geq p-1-q-2. \quad (8.7)$$

Since $q = o(p)$, eventually $\delta(R_1) > 3p/4$. By Turán's theorem, R_1 contains a copy of K_5 . Choose a C_4 and a C_5 inside this fixed K_5 . According as

$$|E(R_1)| \equiv 0, 1, 2 \pmod{3},$$

remove respectively nothing, the C_4 , or the C_5 . Let C denote the removed cycle, allowing $C = \emptyset$, and put

$$H := R_1 - E(C).$$

Every vertex loses either zero or two incident edges when a cycle is removed, so all degrees of H remain even. Moreover, $|E(C)| \equiv |E(R_1)| \pmod{3}$, and hence

$$|E(H)| \equiv 0 \pmod{3}.$$

Therefore H is triangle-divisible. The total loss of degree in passing from R_0 to H is at most four, so

$$\delta(H) \geq p-1-q-4. \quad (8.8)$$

To apply Theorem 2.3 with a fixed parameter, take for example $\varepsilon_0 = 1/100$. Since $q = o(p)$, eventually $q \leq p/20$, and then, for sufficiently large p ,

$$\delta(H) \geq p-1-q-4 \geq 0.91p = (0.9 + \varepsilon_0)|V(H)|. \quad (8.9)$$

Theorem 2.3 now gives an exact triangle decomposition of H . Notice that the decomposition theorem is applied on the original p -vertex set: no vertices are deleted during the correction. The entire correction removes at most $p + 4$ edges.

8.4 Packing size

The combined packing has size

$$\begin{aligned}\nu_3(G) &\geq |F| + \frac{\binom{p}{2} - |F| - O(p)}{3} \\ &= \frac{1}{3} \binom{p}{2} + \frac{2}{3} |F| - O(p) \\ &\geq \frac{1}{3} \binom{p}{2} + \frac{1}{3} \sum_i d_i - O(p + q).\end{aligned}\tag{8.10}$$

Consequently,

$$\begin{aligned}\Phi(G) &\leq \frac{1}{3} \binom{p}{2} + \frac{1}{3} \sum_i d_i + O(p + q) \\ &\leq \frac{1}{3} \binom{p}{2} + \frac{pq}{3} + O(p + q) \\ &= \frac{(p + q)^2}{6} - \frac{p + q^2}{6} + O(p + q).\end{aligned}\tag{8.11}$$

This is at most $n^2/6 + O(n)$.

9. Proof of the Main Theorem

We now prove Theorem 1.1 by contradiction.

Suppose no absolute constant works. For every positive integer k , choose a split graph G_k , minimal in its number n_k of vertices, such that

$$\Phi(G_k) > \frac{n_k^2}{6} + kn_k.\tag{9.1}$$

Necessarily $n_k \rightarrow \infty$. For every $v \in I(G_k)$, minimality gives

$$\Phi(G_k) \leq \Phi(G_k - v) + d(v),$$

and therefore

$$\boxed{d(v) > \frac{2n_k - 1}{6} + k.}\tag{9.2}$$

First, if $q_k \geq 2p_k - 1$, Lemma 5.1 gives

$$\Phi(G_k) \leq \frac{n_k^2}{6} + \frac{p_k}{2},$$

contradicting (9.1) for large k . We may therefore assume $q_k < 2p_k - 1$ and pass to a subsequence according to

$$\alpha_k = \frac{q_k}{p_k} \in [0, 2).$$

9.1 Bulk regime

Suppose that for some $\varepsilon > 0$,

$$\varepsilon \leq \alpha_k \leq 2 - \varepsilon.$$

Theorem 4.2 gives

$$\nu_3^*(G_k) \geq T(G_k) + c_\varepsilon p_k^2 - O(p_k).$$

Haxell–Rödl gives

$$\nu_3(G_k) \geq \nu_3^*(G_k) - o(n_k^2).$$

For sufficiently large k , the quadratic margin dominates the integrality loss, so $\nu_3(G_k) \geq T(G_k)$, contrary to (9.1).

9.2 The endpoint $\alpha_k \rightarrow 0$

This is closed by Section 8, which gives an absolute C_0 such that

$$\Phi(G_k) \leq \frac{n_k^2}{6} + C_0 n_k.$$

For $k > C_0$, this contradicts (9.1).

9.3 The endpoint $\alpha_k \rightarrow 2$

Write

$$q = 2p - s, \quad s = o(p).$$

If $s = O(\sqrt{p})$, inequality (5.3) gives the required linear-error bound. Assume henceforth

$$\sqrt{p} \ll s = o(p). \tag{9.3}$$

For all sufficiently large members of the sequence,

$$p \geq 2304, \quad 6\sqrt{p} \leq s \leq \frac{p}{8}, \quad k \geq 1. \tag{9.4}$$

By (9.2),

$$m_i = |S_i| < \frac{s}{3} + \frac{1}{6} - k \leq \frac{s}{3} - \frac{5}{6}.$$

Since every m_i is an integer, with $m = \max_i m_i$,

$$\boxed{3m \leq s - 3.} \quad (9.5)$$

For $x \in K$, put

$$a_x = |\{i : x \in S_i\}|$$

and

$$D = \sum_{x \in K} a_x (q - a_x). \quad (9.6)$$

Then

$$\sum_{i,j} |S_i \triangle S_j| = 2D. \quad (9.7)$$

High dispersion

Suppose

$$D \geq \frac{qs^2}{12}. \quad (9.8)$$

By (9.5),

$$2p - 3m - 1 \geq 2p - s + 2 = q + 2.$$

Lemma 6.1 and (9.7) therefore give

$$V \geq \frac{q+2}{2}D \geq \frac{q}{2}D. \quad (9.9)$$

The double-factor coefficient in Lemma 5.2 satisfies

$$\boxed{\delta \geq \frac{7}{8}.} \quad (9.10)$$

Indeed, if p is odd then $r_p = p$ and

$$\delta = \frac{p-s}{p} \geq \frac{7}{8},$$

while if p is even then $r_p = p - 1$ and

$$\delta = \frac{p+1-s}{p-1} \geq \frac{7}{8}.$$

Moreover, $S_2 \geq M^2/q$, and (9.5) implies $M/q \leq m \leq (s-3)/3$. Since the resulting parabola is increasing on this interval,

$$\frac{(s-1)M - S_2}{q} \leq \frac{2s(s-3)}{9}. \quad (9.11)$$

Substituting (9.8)–(9.11) into Lemma 5.2 yields

$$\Phi(G) - \frac{n^2}{6} \leq \frac{p}{2} - \frac{5s^2}{288} - \frac{2s}{3}. \quad (9.12)$$

Since $s^2 \geq 36p$,

$$\Phi(G) - \frac{n^2}{6} \leq -\frac{p}{8} - \frac{2s}{3} < 0,$$

a contradiction.

Low dispersion

Suppose

$$D < \frac{qs^2}{12}. \quad (9.13)$$

By (9.7), some j satisfies

$$\sum_i |S_i \triangle S_j| < \frac{s^2}{6}. \quad (9.14)$$

Set

$$R = S_j, \quad \rho = |R|.$$

Then

$$A_R + B_R < \frac{s^2}{6}, \quad \rho \leq m \leq \frac{s-3}{3}. \quad (9.15)$$

We verify the hypotheses of Lemma 7.1 explicitly. Since $s \leq p/8$,

$$q = 2p - s \geq \frac{15p}{8}, \quad b = p - \rho \geq p - \frac{s}{3}.$$

Thus $b \geq 2$, $q \geq r_b$, and $b \geq \chi'(K_\rho)$. Also $r_b \geq b - 1$, so

$$u = q - r_b \leq p - s + \rho + 1.$$

For every i ,

$$2\rho + t_i + 1 \leq 3m + 1 \leq s - 2.$$

Consequently,

$$b - t_i \geq \max\{\rho, u\},$$

and Lemma 7.1 applies.

If $\rho = 0$, then $B_R = 0$. If $\rho \geq 1$, then

$$\theta_R \leq \frac{\rho}{p - \rho} \leq \frac{8s}{23p}.$$

Using $u = q - r_b$, $r_b \leq b$, and $q \geq 15p/8$, we obtain

$$\kappa_R = 1 - 2(1 - \theta_R)\frac{u}{q} \leq \frac{s}{q} + 2\theta_R \leq \frac{5s}{4p}. \quad (9.16)$$

Likewise,

$$\frac{(s - 2\rho - 1)_+}{q} \leq \frac{5s}{4p}. \quad (9.17)$$

By (9.15), the total positive deviation in (7.6) is therefore at most

$$\frac{5s}{4p}(A_R + B_R) < \frac{5s^3}{24p} \leq \frac{5s^2}{192}. \quad (9.18)$$

Meanwhile,

$$\frac{s^2}{6} - s\rho + 2\rho^2 = 2\left(\rho - \frac{s}{4}\right)^2 + \frac{s^2}{24} \geq \frac{s^2}{24}. \quad (9.19)$$

Lemma 7.1 now gives

$$\Phi(G) - \frac{n^2}{6} \leq \frac{p}{2} - \frac{s^2}{64}. \quad (9.20)$$

Since $s^2 \geq 36p$, the right-hand side is at most $-p/16$, again a contradiction.

All possible subsequences are impossible. Theorem 1.1 follows. $\square$

10. Corollaries

10.1 Clique partitions

Corollary 1.2 follows immediately from

$$\text{cp}(G) \leq |E(G)| - 2\nu_3(G).$$

10.2 Sharpness of the leading term

The family

$$K_p \vee \overline{K}_{2p}$$

has $n = 3p$. A factorization of K_p assigned to independent vertices packs every clique edge into a KKI triangle. The resulting triangle-packing expression is

$$|E| - 2\nu_3 = \frac{n^2}{6} + \frac{n}{6}. \quad (10.1)$$

The Erdős–Ordman–Zalcstein clique-partition construction on this family shows that the coefficient $1/6$ cannot be improved.

10.3 Localization of difficult sequences

Proposition 1.3 records the localization statement supplied by the proof. It should be understood as a structural description of a hypothetical failure of every linear-error estimate, rather than as an additional extremal theorem after Theorem 1.1. The bulk margin excludes ratios bounded away from 0 and 2, the sparse construction excludes $|I|/|K| \rightarrow 0$, and averaged factorization excludes the short corridor near $|I| = 2|K|$. The only remaining location is the mesoscopic corridor

$$|I| = 2|K| - s, \quad \sqrt{|K|} \ll s = o(|K|),$$

which is then eliminated by the dispersion dichotomy.

10.4 Exact common-profile fractional value

Theorem 3.1 is independently useful: it gives the exact fractional triangle packing for every split graph in which all independent vertices have one common neighborhood, including the regime with too few independent vertices to color all neighborhood edges integrally. We state it as a stand-alone corollary.

Corollary 10.4a — Exact common-neighborhood triangle packing

For all integers $p \geq 3$, $q \geq 1$ and $0 \leq d \leq p$, the graph $H(p, q, d)$ (clique of order p ; q independent vertices, each with the same neighborhood of size d) satisfies

$$\nu_3^*(H(p, q, d)) = F(p, q, d) = \min \left\{ \frac{\binom{p}{2} + qd}{3}, \binom{d}{2} + \binom{p-d}{2}, \binom{d}{2} + \frac{d(p-d) + \binom{p-d}{2}}{3} \right\}.$$

The same identity holds for $q = 0$, where $H(p, 0, d) = K_p$ and $\nu_3^*(K_p) = \binom{p}{2}/3$. This is an exact closed form for the fractional triangle-packing number of an entire one-parameter family, valid for every d , including the small- q regime where no integral factorization covers the neighborhood; it may be of independent interest outside the clique-partition problem.

Corollary 10.4b — Threshold graphs

Since $K_p \vee \overline{K}_{2p}$ is a threshold graph and every threshold graph is split, Theorem 1.1 gives, in particular, the linear-error bound $\text{cp}(G) \leq n^2/6 + Cn$ for all threshold graphs, with the same sharp leading constant witnessed inside that subclass.

10.5 Localized effectivity near $q = 2p$

Proposition 10.5 — Explicit corridor bounds

Let $q = 2p - s$ and $n = p + q$.

1. If

$$p \geq 36, \quad 0 \leq s \leq 6\sqrt{p},$$

then

$$\boxed{\Phi(G) \leq \frac{n^2}{6} + 2n.} \quad (10.2)$$

2. If

$$p \geq 2304, \quad 6\sqrt{p} \leq s \leq \frac{p}{8},$$

and every $v \in I$ satisfies

$$d(v) > \frac{2n-1}{6} + 1,$$

then

$$\boxed{\Phi(G) \leq \frac{n^2}{6}.} \quad (10.3)$$

Consequently, for every $C \geq 2$, a minimum-order counterexample to

$$\Phi(G) \leq \frac{n^2}{6} + Cn$$

cannot satisfy

$$p \geq 2304, \quad 0 \leq s \leq \frac{p}{8}.$$

Proof

For the short corridor, inequality (5.3) and $s^2 \leq 36p$ give

$$\Phi(G) - \frac{n^2}{6} \leq \frac{7p}{2} + \frac{1}{4} \leq 2n,$$

because $p \geq 36$ implies $s \leq p$ and hence $n = 3p - s \geq 2p$.

The mesoscopic assertion is exactly the quantitative high/low-dispersion calculation in Section 9.3. The final consequence follows from the degree inequality obtained by deleting an independent vertex from a minimum-order counterexample. $\square$

11. Discussion

11.1 From fractional asymptotics to a linear-error integral theorem

The central difficulty is that the general theorem

$$\nu_3^*(G) - \nu_3(G) = o(n^2)$$

has no linear rate. The present proof avoids demanding one uniform rounding mechanism.

- In the bulk, the common-profile LP produces quadratic slack.
- Near $q = 2p$, explicit factorization structures replace general rounding.
- Near $q = 0$, the residual clique becomes dense enough for exact decomposition.

This regime-dependent strategy is weaker than a universal linear integrality-gap theorem but sufficient for the extremal clique-partition problem. The packing-versus-covering theme for triangles is classical; Tuza’s conjecture and its fractional forms, for which Krivelevich [14] proved two fractional variants and an integral special case, provide the closest general context, though they are not used in the present proof.

11.2 Why residual maximum degree is not the right invariant

Early exploratory work attempted to prove that an optimal packing leaves a clique residual of bounded maximum degree. Numerical behavior depended strongly on the packing algorithm, and no such property is needed in the final proof.

The effective invariants are instead:

- the common-profile fractional margin;
- the variance term S_2 ;
- the polarization energy V ;
- the centered deviations A_R, B_R ;
- divisibility of the dense residual.

11.3 Localized effectivity and remaining non-effectivity

Proposition 10.5 shows that the entire near-extremal corridor treated by factorization, polarization, and shifted-center completion is quantitatively effective. One may take

$$C_{\text{corr}} = 2, \quad p_0 = 2304, \quad s_0(p) = 6\sqrt{p}, \quad \eta_0 = \frac{1}{8}.$$

The global constant C remains non-effective for two independent reasons.

1. The Haxell–Rödl theorem is used through an asymptotic $o(n^2)$ statement in the bulk.
2. Dense triangle decomposition theorems are invoked with unspecified sufficiently-large thresholds in the sparse-independent regime.

Thus no hidden non-effectivity remains in Sections 5–7; it comes only from the two external asymptotic inputs used elsewhere in the proof. After Appendix D, Sections 5–7, the high/low-dispersion argument of Section 9.3, and Proposition 10.5 are self-contained and effective. Proposition 10.5 uses no instance of Theorem 2.1 or Theorem 2.3; its only external background is the standard edge coloring of complete graphs together with the list edge coloring theorem proved in Appendix

D. Dirac’s and Turán’s theorems belong to the separate sparse-independent argument of Section 8.

11.4 Relationship with the fractional extremal theorem

The complete-split extremal theorem determines the global fractional maximum of

$$|E(G)| - 2\nu_3^*(G)$$

over both split and chordal graphs. The common-profile LP in the present paper serves a different purpose: it is a profile-sensitive local lower bound for the fractionally cloned graph, and its quantitative slack absorbs integrality loss away from the extremal corridor.

Thus the global fractional extremal calculation and the local common-profile calculation are compatible but logically distinct. The present proof is self-contained apart from the external theorems stated in Section 2.

Remark (cloning terminology). Lemma 4.1 and the clone-symmetrization step of Paper II use the same underlying replication operation in different roles. Paper II uses pairwise cloning to move a chordal graph toward a complete-split terminal graph. Lemma 4.1 uses q -fold profile replication on the cover side to compare an arbitrary profile with a common-profile graph. The established term *fractional cloning* is retained for the present inequality. Paper IV uses the term *clone step* for a different operation, the vertex-copy substitution that replaces one neighbourhood rather than replicating a profile.

11.5 Relationship with the chordal linear-error problem

The complete-split reduction settles the asymptotically sharp chordal bound

$$\text{cp}(G) \leq \left(\frac{1}{6} + o(1)\right) n^2.$$

The stronger estimate

$$\text{cp}(G) \leq \frac{n^2}{6} + O(n)$$

remains open for general chordal graphs. The mechanisms developed here suggest a concrete three-part program for that problem.

1. **Stability of the fractional extremal theorem.** Upgrade the exact chordal fractional theorem of Paper II to a stability statement: every chordal graph whose value is within δn^2 of the fractional maximum is within εn^2 edge edits of a maximizing complete split graph. Away from near-extremality, the quadratic fractional slack absorbs the general integrality loss exactly as in Section 4.
2. **Robust corridor mechanisms.** Extend the factorization, polarization, and shifted-center arguments of Sections 5–7 from split graphs to εn^2 -perturbations of complete split graphs, so that they apply to the near-extremal graphs produced by stability.
3. **Clique-tree assembly.** Patch the local estimates across the clique tree by an edge-ownership ledger that preserves separator capacities, in the spirit of shifted-center completion, so that no edge is charged twice.

A chordal linear-error proof would still have to preserve ownership and separator capacities across the clique tree; the ledger of step 3 is designed for exactly that purpose.

Paper V of the series, in preparation, investigates this program. Its intended clique-separator recursion uses the split theorem proved here as the base case and is designed so that the chordal regimes mirror the corresponding ingredients of Sections 3–9: common-profile evaluation, fractional cloning, factorization, and corridor analysis. Until that companion argument is complete, this paragraph records the proposed interface rather than a theorem of the present paper.

11.6 Formal-verification perimeter and current status

Papers I and II of the series have frozen Lean 4 / Mathlib releases. The present paper is being prepared under the same release protocol. The author-supplied report records a Mathlib v4.28.0 build comprising 8,057 jobs, zero errors, and zero `sorry`; the public commit and an independent reproduction remain pending.

The formal development separates two layers.

- **Layer X** contains exactly the two external asymptotic inputs, Theorems 2.1 and 2.3, retained as named axioms **AX1** and **AX2** with their hypotheses stated explicitly. In the formalization **AX1** is stated on the cover side ($\tau_3^* - \nu_3 = o(n^2)$); the proved weak-duality inequality $\nu_3^* \leq \tau_3^*$ is sufficient for the bulk deduction used here. No additional strong-duality assumption is required for that step.
- **Layer E** contains the finite mathematical core recorded by the supplied report as proved from first principles and machine-checked: Theorems 3.1 and 4.2; Lemmas 4.1–7.1; the divisibility correction of Appendix B; the list edge coloring development of Appendix D; the quantitative chains of Sections 9.3 and 10.5; Proposition 10.5; and the final assembly of Theorem 1.1. Two reusable classical results absent from Mathlib — Dirac’s Hamiltonicity theorem and a near-perfect matching from minimum degree — are also proved from scratch and are axiom-clean.

The supplied report records the following axiom footprints: Theorem 1.1 and Corollary 1.2 carry `propext`, `Classical.choice`, `Quot.sound`, **AX1**, and **AX2**; the bulk node carries **AX1** in addition to the standard triple; and Proposition 10.5 together with the elementary lemmas and the divisibility correction carry only the standard triple. The same report lists both **AX1** and **AX2** for the sparse-regime node, although the manuscript proof in Section 8 visibly invokes only Theorem 2.3 (**AX2**) among the two named asymptotic inputs. Whether **AX1** is a genuine transitive dependency or an avoidable import remains a release-gate item to be reconciled against the frozen axiom report. The supplied report records no `sorryAx`, `admit`, or `native_decide`, and identifies **AX1** and **AX2** as the only project axiom declarations.

Table 2. Reported formalization perimeter. Layer X contains the two external inputs retained as named axioms; Layer E is the finite core recorded as machine-checked relative to those axioms. “Lean report: proved” refers to the author-supplied build and axiom report; independent reproduction and human peer review remain pending.

Layer	Node	Content	Status
X	AX1	Haxell–Rödl/Yuster: $\tau_3^* - \nu_3 = o(n^2)$	Named axiom (external)
X	AX2	Dross–BKLO: triangle-divisible, $\delta \geq (0.9 + \varepsilon)n \Rightarrow$ triangle decomposition	Named axiom (external)

Layer	Node	Content	Status
E	Thm 3.1	common-profile exact fractional value $F(p, q, d)$	Lean report: proved
E	Thm 4.2	unified fractional margin	Lean report: proved
E	Lemmas 4.1–7.1	cloning, factorization, polarization, shifted-center	Lean report: proved
E	App. B	divisibility correction (Dirac ordering + parity)	Lean report: proved
E	App. D	list edge coloring (Galvin/König, self-contained)	Lean report: proved
E	Prop 10.5	explicit corridor bounds (effective, unconditional)	Lean report: proved
E	Bulk node (§9.1)	fractional-to-integral in the bulk	Lean report: proved, modulo AX1
E	Sparse node (§8)	dense residual decomposition	Lean report: proved, reported modulo AX1 , AX2 ; manuscript proof visibly uses AX2 ; dependency reconciliation pending
E	Thm 1.1	main theorem (assembly)	Lean report: proved, modulo AX1, AX2

No computation or regression script is a premise of the proof. The exact-arithmetic and integer-programming checks are supplementary audits. The supplied audit summary reports, under one counting convention, 46,390 exact-arithmetic and LP checks and 91 exact ILP checks, for a total of 46,481 checks, with no reported discrepancy.

The public-preprint status line will be tied to the frozen release commit, the axiom report, and the Lean toolchain and manifest. Independent reproduction of the build from that public commit, human peer review, and the prior-art assessment remain pending; on those grounds the wording “formally verified” is reserved for the frozen release.

12. Potential Uses and Future Directions

12.1 Universal linear integrality gap

Problem 12.1

Is there an absolute constant C such that

$$\nu_3^*(G) - \nu_3(G) \leq C|V(G)|$$

for every split graph G ?

The present paper supplies supporting structures near the extremal corridor but does not resolve the bulk regime without quadratic slack.

12.2 Effective global constants

Problem 12.2

Find an explicit global constant C in Theorem 1.1.

The near-extremal corridor is already effective by Proposition 10.5. A fully quantitative global theorem therefore requires only explicit rates in fractional-to-integral packing, or a structured replacement for Haxell–Rödl in the bulk, together with effective dense-decomposition thresholds in the sparse-independent regime.

12.3 Algorithmic packing

The proof contains polynomially implementable pieces:

- complete-graph factorizations;
- weighted selection of reserved vertices;
- bipartite list edge coloring;
- matching extraction;
- dense decomposition algorithms implicit in iterative absorption.

It is natural to seek a polynomial-time algorithm that outputs a clique partition of size

$$\frac{n^2}{6} + O(n)$$

with an explicit constant. The factorization, list-coloring, and matching steps of Sections 5–7 are individually polynomial; assembling them into a single approximation algorithm with a proven guarantee is left to future work.

Corollary 12.3 — Effective corridor algorithm

In the near-extremal corridor covered by Proposition 10.5 (explicit p_0, s_0), the proof is constructive: the one- and double-factorizations of K_p , the reserved-vertex selection, the list edge coloring (Appendix D, whose kernel/stable-pairing proof is itself an algorithm), and the matching extraction all run in polynomial time. Hence there is a deterministic polynomial-time algorithm that, for split graphs in the effective corridor, outputs a clique partition of size $n^2/6 + 2n$ with the explicit constant of Proposition 10.5. Outside the corridor the guarantee currently relies on the non-constructive bulk input (Theorem 2.1); a fully effective global algorithm is left to future work.

12.4 Stability and extremal classification

Problem 12.4

Characterize split graphs satisfying

$$\text{cp}(G) \geq \frac{n^2}{6} - o(n^2).$$

The proof suggests that near-extremal graphs should have $|I| = (2 + o(1))|K|$ and low-dispersion absence profiles close to a common center.

Two pieces of evidence support a Simonovits-type stability statement. First, an exhaustive enumeration at $n = 9$ (all split graphs with $|K| = 3$) shows that graphs within an additive constant of the maximum are within a bounded edit distance of the complete-split family, while

genuinely dispersed profiles fall a linear amount below the maximum. Second, the fractional margin of Theorem 4.2 already yields a level-set bound: away from near-extremality the quadratic slack $\mu(\alpha)p^2$ is strictly positive, so any near-maximizer must have α near the extremal ratio and small profile dispersion. A full stability theorem would still require quantifying the gain of a single symmetrization step and controlling the directions along which Φ^* is flat; we leave it as an open problem, most naturally phrased in the cut (graphon) metric to absorb the choice of nearest complete-split graph.

12.5 Higher-clique linear-error packing

The complete-split reduction determines the fractional extremal problem and the first-order integral asymptotics for fixed K_r -and-edge partitions. The natural higher-clique analogue of the present paper is the following linear-error question.

For fixed $r \geq 3$, does every split graph satisfy

$$\pi_r(G) \leq \frac{r-1}{4r}n^2 + O_r(n),$$

where $\pi_r(G)$ is the minimum number of parts in an edge partition into copies of K_r and single edges?

The common-profile and cloning ideas suggest one possible approach, but the relevant local orbit programs and integral rounding mechanisms become substantially more complicated. Design and decomposition methods such as those developed by Keevash [15] and the regular-slice framework of Allen, Böttcher, Cooley, and Mycroft [16] may be useful tools; they are not ingredients of the present proof.

12.6 Chordal and clique-tree applications

Shifted-center completion was designed to separate owned clique edges from protected interface edges. A parametric version may be useful in a future chordal linear-error theorem, where separators are shared between neighboring maximal cliques.

12.7 Reusable proof principle

The paper illustrates a broader strategy:

Use exact symmetric LPs to create slack away from extremality, then reserve explicit combinatorial rounding only for the corridor where the slack vanishes.

This may apply to other structured packing-covering problems in which a general regularity theorem is too weak at second order.

13. Reproducibility

The intended working release package for this draft is organized as follows.

```
01_MANUSCRIPT/
  PAPER_III_split_lineal_v0.9.12_editorial_review_en.md
02_PROOF_CONTROL/
  LEDGER.md
  LEDGER_INCREMENTAL_formalization_draft.md
  AUDIT_PAPER_C.md
```

```

03_FORMALIZATION/
  lean/                                (Lean 4 / Mathlib v4.28.0 sources)
  lakefile.toml
  lean-toolchain
  lake-manifest.json
  BUILD_STATUS.md                     (temporary until the release build is frozen)
04_SUPPLEMENTARY_REGRESSION/
  audit_c_fast.py
  audit_c_ilp.py
  audit_c_ilp_results.txt

```

The manuscript is the mathematical source of truth. `LEDGER.md` is the authoritative dependency specification for the Lean development; the incremental ledger records only the transition from v0.9.1 to this draft. The final release package will add the frozen commit, build log, axiom report, and SHA-256 manifest. Before release, the script index in this section must be reconciled with the more detailed audit inventory in Appendix C; no script name or check count should be treated as frozen until that reconciliation is complete.

The proof itself is contained in the manuscript. The common-profile optimization is proved in Theorem 3.1; the fractional margin is proved in Theorem 4.2 and Appendix A; the factorization, polarization, and shifted-center inequalities are proved in Lemmas 5.1–7.1; the divisibility correction is proved in Section 8.3 and Appendix B; and the explicit near-extremal constants are proved algebraically in Sections 9.3 and 10.5.

The scripts listed in the working release package are supplementary regression tests for the explicit corridor constants. Their outputs are not used as hypotheses, reductions, or proof steps anywhere in the manuscript. The script inventory remains subject to reconciliation with Appendix C before the release package is frozen. Historical exploratory computations are not part of the logical proof package.

The public release will be made through the series repository at a frozen tag and commit. Manuscripts, Markdown, LaTeX, PDFs, and related written materials are intended to be distributed under CC BY-NC 4.0. Author-owned verification and regression code is intended to be distributed under the PolyForm Noncommercial License 1.0.0; commercial software licensing is available from the author. Third-party dependencies retain their own licences.

Open release gates. Before public release, the package index and Appendix C must be reconciled; the Lean toolchain, manifest, build log, and axiom report must be frozen at the public commit; the reported computational totals must be regenerated from the packaged scripts; and an independent build reproduction, human peer review, and prior-art review remain pending.

Appendix A. Algebra of the Fractional Margin

Let $x = d/p$ and $\alpha = q/p$. Ignoring the exact linear terms temporarily, the three normalized branches of F are

$$f_1(x) = \frac{1}{6} + \frac{\alpha x}{3},$$

$$f_2(x) = x^2 - x + \frac{1}{2},$$

and

$$f_3(x) = \frac{1}{6} + \frac{x^2}{3}.$$

Subtracting the affine target contribution gives minima

$$\min_x \left(f_1(x) - \frac{\alpha x}{2} - C_\alpha \right) = \frac{\alpha^2}{12},$$

$$\min_x \left(f_2(x) - \frac{\alpha x}{2} - C_\alpha \right) = \frac{(2 - \alpha)^2}{48},$$

and

$$\min_x \left(f_3(x) - \frac{\alpha x}{2} - C_\alpha \right) = \begin{cases} \alpha(8 - 5\alpha)/48, & \alpha \leq 4/3, \\ (2 - \alpha)^2/12, & \alpha \geq 4/3. \end{cases}$$

The third is dominated by the lower envelope of the first two. Restoring the exact $-p/2$ contribution in each pointwise estimate and averaging produces the $-p/4$ term in Theorem 4.2.

Appendix B. Divisibility Correction

Let $P = x_1 \cdots x_p$ be a path and let $O \subseteq V(P)$ have even cardinality. Define

$$J = \{x_j x_{j+1} : |O \cap \{x_1, \dots, x_j\}| \text{ is odd}\}.$$

Every internal vertex x_j changes prefix parity precisely when $x_j \in O$. Therefore

$$\text{Odd}(J) = O.$$

Also $|E(J)| \leq p - 1$ and $\Delta(J) \leq 2$.

After parity correction, the minimum degree has fallen by at most two. In the sparse-independent regime it is still greater than $3p/4$ for all sufficiently large p , so Turán's theorem supplies a K_5 . Deleting a C_4 inside that K_5 changes the edge count by 1 (mod 3), while deleting a C_5 changes it by 2 (mod 3). Every affected degree changes by two, so parity is preserved. The total degree loss from the path subgraph and the correcting cycle is at most four. Thus, when $q = o(p)$, the final graph has minimum degree at least $p - 1 - q - 4$, which eventually exceeds $(0.9 + \varepsilon_0)p$ for any fixed $0 < \varepsilon_0 < 0.1$.

Appendix C. Computational Audits

C.1 Exact common-profile LP

`verify_common_profile_lp.py` enumerates the vertices of the symmetrized cover polyhedron using exact rational Gaussian elimination and compares the optimum with (3.5).

C.2 Exact fractional margin

`verify_fractional_margin.py` checks Theorem 4.2 for

$$3 \leq p \leq 80, \quad 1 \leq q \leq 2p, \quad 0 \leq d \leq p$$

using exact rational arithmetic.

C.3 Small ILP audits

`verify_factor_rounding.py` and `verify_shifted_center.py` compute exact integral triangle packings on small instances using a binary edge-capacity ILP. They verify the stated upper bounds on $\Phi(G)$.

C.4 Polarization

`verify_polarization.py` performs exhaustive small-profile and randomized exact-integer checks of Lemma 6.1.

C.5 Divisibility

`verify_divisibility_correction.py` exhaustively verifies the path parity construction through order eighteen, checks the fixed minimum-degree threshold exactly, and follows the full parity and modulo-three correction on randomized dense residuals.

Appendix D. Self-Contained List Edge Coloring

This appendix proves the list edge coloring theorem used in Section 7.2 (stated as Theorem 2.2), so that the corridor argument depends on no external coloring theorem. The result is the maximum-degree case of Galvin’s theorem [10]; the gain graph of Section 7.2 is simple and bipartite, which is the case proved here.

D.1 Kernels

A **kernel** of a digraph D is an independent set K of vertices such that every vertex outside K has an out-neighbor in K . D is **kernel-perfect** if every induced subdigraph has a kernel. A kernel of a nonempty digraph is nonempty.

Lemma D.1 — Kernel coloring lemma

Let D be a kernel-perfect digraph in which every vertex v has a list $L(v)$ with $|L(v)| \geq d_D^+(v) + 1$. Then the underlying graph of D has a proper coloring choosing each color from the corresponding list.

Proof

Pick any color c appearing in some list, let $S = \{v : c \in L(v)\}$, and let K be a kernel of $D[S]$. Color every vertex of K with c ; this is proper on K because K is independent. Delete K from D and delete c from the list of every vertex of $S \setminus K$. Every $v \in S \setminus K$ lost one color but also at least one out-neighbor, namely its out-neighbor in K , so the hypothesis $|L(v)| \geq d^+(v) + 1$ is preserved; vertices outside S keep their lists while their out-degrees do not increase. Repeat. Every uncolored vertex always has a nonempty list, and each round with $S \neq \emptyset$ colors the nonempty set K , so every vertex is eventually colored. $\square$

D.2 Stable pairings

Let B be a bipartite graph with parts U and R in which every vertex z has a linear preference order $>_z$ on its incident edges. A matching $M \subseteq E(B)$ is **stable** if every edge $f \notin M$ has an endpoint z that is covered by an edge $e \in M$ with $e >_z f$.

Lemma D.2 — Gale–Shapley [11]

For every system of preferences, a stable matching exists.

Proof

Run deferred acceptance. While some $u \in U$ is unmatched and has not yet proposed along all of its edges, let u propose along its most preferred untried edge, say f with endpoint $r \in R$. If r is unmatched, or prefers f to its currently held edge, then r accepts f , releasing its previous partner, if any; otherwise r rejects f . Every edge is proposed at most once, so the process terminates, and it terminates in a matching M .

Let $f = ur \notin M$. Suppose first that u proposed along f at some point. Then r either rejected f immediately or accepted it and later released it; in both cases, at that moment r held or received an edge it strictly prefers to f . Since the edge held by r only improves during the process, the final edge $e \in M$ at r satisfies $e >_r f$. Suppose instead that u never proposed along f . An unmatched vertex of U keeps proposing while untried edges remain, so at termination u is matched, say to $e \in M$; and since u proposes in decreasing order of preference and never reached f , we have $e >_u f$. In both cases f has an endpoint whose matched edge is preferred to f . $\square$

D.3 The coloring theorem

Theorem D.3 — Galvin, maximum-degree case

Let B be a simple bipartite graph with maximum degree Δ , and let every edge e carry a list $L(e)$ with $|L(e)| \geq \Delta$. Then B has a proper edge coloring choosing each color from the corresponding list.

Proof

Step 1: a proper Δ -edge-coloring. By König’s edge coloring theorem [12], B has a proper edge coloring $\varphi : E(B) \rightarrow \{1, \dots, \Delta\}$. For completeness: induct on the number of edges. Remove an edge $e = ur$ and color the rest. At most $\Delta - 1$ colors appear at u and at most $\Delta - 1$ at r , so u misses a color α and r misses a color β . If they miss a common color, give it to e . Otherwise consider the maximal path P starting at u whose edges are colored alternately β, α . The path P cannot end at r : since r misses β , such a path would end with an α -edge and hence have even length, while every path between the adjacent vertices u and r of a bipartite graph has odd length. Swap the colors α and β along P ; by maximality of P the coloring stays proper, the vertex r is untouched, and u now misses β . Give β to e .

Step 2: the orientation. Define preferences from φ : every $u \in U$ prefers its incident edges with **higher** φ -color, every $r \in R$ prefers **lower** φ -color. Define a digraph D on the vertex set $E(B)$: for distinct edges e, f sharing an endpoint z , put an arc $e \rightarrow f$ exactly when $f >_z e$.

Out-degrees: let $e = ur$ with $\varphi(e) = c$. The out-neighbors of e are the edges at u with color greater than c — at most $\Delta - c$, since the colors at u are pairwise distinct — and the edges at r with color smaller than c — at most $c - 1$. Hence

$$d_D^+(e) \leq (\Delta - c) + (c - 1) = \Delta - 1.$$

Step 3: kernel-perfectness. An induced subdigraph of D is $D[S]$ for a set S of edges of B . A kernel of $D[S]$ is exactly a stable matching of the subgraph $(V(B), S)$ with the inherited preferences: independence in D means that no two kernel edges share an endpoint, because two edges at a common vertex are always joined by an arc; and the domination condition asks that every $f \in S \setminus K$ have an arc into K , that is, an endpoint z and an edge $e \in K$ at z with $e >_z f$, which is stability. Lemma D.2 provides a stable matching, so D is kernel-perfect.

Step 4: conclusion. Every edge has $|L(e)| \geq \Delta \geq d_D^+(e) + 1$, so Lemma D.1 yields a proper coloring of the underlying graph of D from the lists — that is, a proper list edge coloring of B . $\square$

Remark D.4 — Application

The gain graph of Section 7.2 is simple and bipartite, and hypothesis (7.2) bounds every list by $b - t_i \geq \max\{\rho, u\} \geq \Delta$ of the gain graph. Theorem D.3 therefore applies verbatim. The multigraph version of Galvin’s theorem holds as well [10] and is not needed here.

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